# Juan Carlos Delgado-Pivaral

626-422-1132 | jdelgadopivaral@college.harvard.edu | LinkedIn | Portfolio

## EDUCATION:

**Harvard College (GPA: 3.735)** Expected May 2029

- Bachelor of Science (S.B.) Candidate in Mechanical Engineering

## TECHNICAL SKILLS:

**Software & Design:** SolidWorks, AutoCAD, Onshape, Technical Drawings, Python, LaTeX, Design for Manufacturing
**Manufacturing:** CNC Machining, Manual Milling, Lathe, 3D Printing, Laser Cutting, Mold Fabrication, CAM
**Languages:** English (Native), Spanish (Native)

## ENGINEERING EXPERIENCE:

**Westin Automotive Products** San Dimas, CA
*Product Development Engineering Intern* *June 2026 – August 2026*

- Designed automotive lighting assemblies in SolidWorks and managed their transition from prototype development to manufacturing release.
- Developed prototypes alongside machinists and conducted vehicle fit-ups to evaluate component integration, identify design issues, and refine mechanical designs.
- Coordinated directly with an overseas manufacturer to support product releases for production and prepared detailed technical documentation for mechanical component replacement.

**Harvard Undergraduate Automotive Society – Formula SAE** Cambridge, MA
*Mechanical Team | Steering Sub-team* *September 2025 – Present*

- Designed steering components and assemblies in SolidWorks, supporting vehicle development through CAD modeling, prototyping, and mechanical integration.
- Evaluating impact attenuator designs and developing an anti-intrusion plate mounting assembly in SolidWorks, considering chassis integration and Formula SAE requirements.

**Harvard Undergraduate Robotics Club – University Rover Challenge** Cambridge, MA
*Mechanical Engineer | Science Team* *September 2026 – Present*

- Serve as the designated mechanical engineer for both science subteams, translating scientific objectives into mechanical designs and coordinating system integration.
- Designing a rover-mounted drilling and auger system for subsurface soil collection, accounting for chassis geometry, mounting configurations, and required drilling depths.
- Designing a soil caching system to separate shallow and deep samples, incorporating removable storage and a retrieval mechanism meeting a five-minute requirement.

## PROJECTS:

**Battle Robot: Mechanical Design & Manufacturing**

- Designed drivetrain, gearbox, wheel, and structural components in SolidWorks, incorporating manufacturing constraints, component integration, and mechanical assembly requirements.
- Manufactured and assembled custom robot components through machining, laser cutting, 3D printing, and molding, refining designs through prototyping, fit-ups, and mechanical testing.
- Collaborated on robot integration and performance improvements, earning 1st place in the final competition.

## LEADERSHIP:

**SHPE (Society of Hispanic Professional Engineers)** Cambridge, MA
*Co-Director of Outreach* *September 2025 – Present*

- Lead chapter outreach through newsletters, social media, and event coordination, promoting professional opportunities and strengthening engagement with students and partner organizations.

## ADDITIONAL EXPERIENCE:

**Pauline's Food Service (Family-Run)** Los Angeles, CA
*Cook, Caterer, Translator* *January 2020 – Present*

- Coordinate catering for 40–500+ guests, provide bilingual translation, and trained 5+ new employees.

**Covina High School** Covina, CA
*Multi-Subject Tutor* *August 2024 – July 2025*

- Tutored students in mathematics through Calculus BC, physics, and chemistry in individual and group sessions.

## EXTRACURRICULAR INVOLVEMENT:

**Harvard Men's Club Soccer** Cambridge, MA

- Selected to Harvard Men's Club Soccer A Team, competing in intercollegiate matches and tournaments.